

Data Analysis & Probability



Mean, median, mode and range

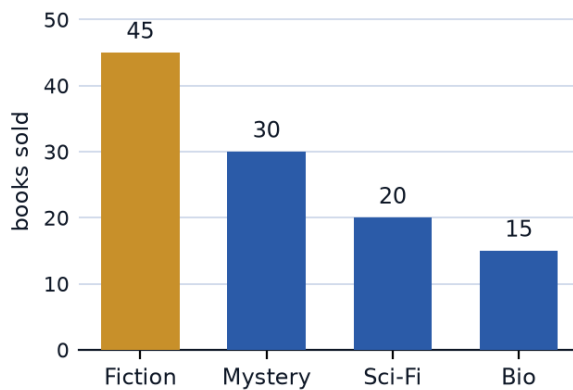
- 1 Find the mean of 6, 9, 12, 15, 18.
 - 2 Ages of 7 students: 13, 15, 13, 16, 14, 13, 17. Find the mean, median, and mode.
 - 3 Find the range and mode of the data set 8, 3, 8, 15, 8, 22, 3.
 - 4 A data set has mean 18 and 6 values. Find the sum of the values.
 - 5 The data set 10, 12, 11, 13, 50 contains an outlier. Find the mean before and after removing 50.
 - 6 A data set of 5 values has mean 20. Four of the values are 15, 18, 22, 25. Find the fifth value.
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Reading tables

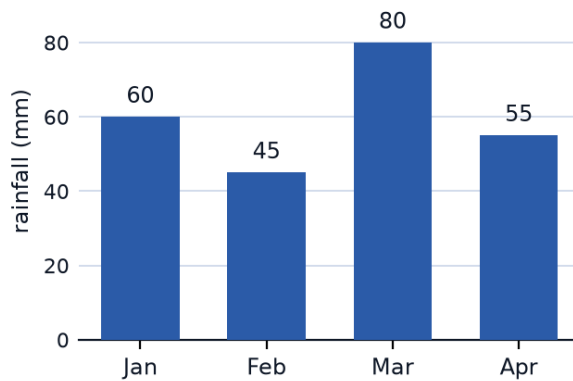
- 7 A survey of 50 people's favorite fruit: Apple 15, Banana 12, Orange 10, Mango 13. What fraction of people chose banana?
 - 8 Using the same survey (Apple 15, Banana 12, Orange 10, Mango 13, out of 50), what percent chose either apple or mango?
 - 9 A table shows quiz scores: score 5 (3 students), score 7 (5), score 8 (8), score 9 (4), score 10 (2). Find the mode score and the total number of students.
 - 10 Using the same quiz data, find the mean score to two decimal places.
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Reading bar charts

- 11 The bar chart shows books sold by genre in one week. Find the total books sold and the modal (most common) genre.



- 12 Using the same book-sales data, what percent of total books sold were mystery novels, to the nearest percent?
- 13 The bar chart shows monthly rainfall in mm. Find the mean monthly rainfall.



- 14 Using the same rainfall data (Jan 60, Feb 45, Mar 80, Apr 55 mm), find the percent decrease from March to April.

Basic probability

- 15 A bag holds 4 red and 6 blue marbles. Find $P(\text{red})$.
- 16 A fair six-sided die is rolled. Find $P(\text{a number greater than 4})$.
- 17 A card is drawn from a standard deck of 52 cards. Find $P(\text{a heart})$.
- 18 A spinner has 8 equal sections numbered 1 to 8. Find $P(\text{an even number})$.
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Combined and complementary events

- 19 A bag holds 3 red, 5 blue and 2 green marbles. Find $P(\text{not red})$.
- 20 Two fair coins are tossed. Find $P(\text{both land heads})$.
- 21 A die is rolled twice. Find $P(\text{both rolls show a 6})$.
- 22 A bag holds 4 white and 6 black balls. One ball is drawn and not replaced, then a second is drawn. Find $P(\text{both white})$.
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Weighted averages

- 23 A class of 9 students has an average score of 80. A 10th student joins with a score of 100. Find the new average for all 10 students.
- 24 Class A has 20 students with mean score 75; Class B has 30 students with mean score 85. Find the overall mean of all 50 students.
- 25 A student's final grade is 40% homework average (score 90) and 60% exam average (score 70). Find the weighted final grade.
- 26 A survey combines two groups: 15 people with average age 24, and 25 people with average age 32. Find the overall average age of all 40 people.
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Comparing data sets

- 27 Data set A: 10, 12, 14, 16, 18. Data set B: 8, 14, 14, 14, 20. Both have the same mean. Which has the greater range?
- 28 Team X scored 22, 25, 28, 25 points in 4 games. Team Y scored 20, 20, 35, 25. Both teams have the same mean score. Which team's scores are more consistent (smaller range)?
- 29 Class 1 has 18 students with mean score 68; Class 2 has 22 students with mean score 74. Find the combined mean of all 40 students, to one decimal place.
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Word problems

- 30** A teacher records the number of siblings for 10 students: 1, 2, 0, 1, 3, 2, 1, 0, 2, 1. Find the mean, median, and mode.
- 31** A company's weekly sales (in thousands of riyals) over 5 weeks: 40, 55, 35, 60, 50. Find the mean weekly sales, and state how many weeks had sales above the mean.
- 32** A jar contains 5 red, 3 blue and 2 yellow candies. A candy is picked and eaten, then another is picked. Find $P(\text{first red, then blue})$.
- 33** In a survey of 200 students, 120 prefer math and the rest prefer science. Find the probability, as a percent, that a randomly chosen student prefers science.